University of Cologne

Institute of Physics II - Practical Course M

M2.10: Ultrafast Transient Grating Spectroscopy



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1 Introduction

In this experiment the ambipolar mobility of photo-excited charge carriers in the semiconductor Gallium Arsenide (GaAs) is measured at room temperature with a highly advanced state-of-the-art spectroscopic technique and the principles of ultrafast time-resolved spectroscopy are discussed.
At first, sub-picosecond laser pulses are characterized by measuring their duration via an optical autocorrelation. In the second part of the lab course experiment, pulses from the same laser source are employed to induce a spatially modulated excitation pattern, the Transient Grating, at the surface of the GaAs sample and to observe its spatio-temporal evolution. In this way, it is possible to study both, the recombinative and diffusive contributions to the relaxation processes. Finally, the ambipolar diffusivity and mobility of the photo-excited charge carriers can be extracted.

To measure dynamical processes on ultrashort time-scales requires to use a measurement tool which is able to resolve the dynamical process of interest. In order to reach the resolution needed, the measurement tool has to be faster or shorter in time than the observed process itself. Take a simple example, if you want to record the movement of a climbing snail on a tree, one only needs a dummy point-and-shoot camera and press its taking button every second to record a lot of pictures. These separate pictures are sufficient to follow and tell the movement of the slow snail. The time resolution in this case can reach hundreds of milliseconds, dependent on how fast you can press the button and release it. However, if one wants to record a flying bird, the dummy camera and the movement of your fingers apparently is too slow and hence the recorded image is blurred. To overcome such an issue, an electronic fast switching special camera is required to record images continuously, i.e. filming. Electronical devices and special fast cameras can reach resolutions up to microsecond (10^{-6}s) time scales. In solid state materials, the events and processes that are of interest, occur on usually very short timescales far below the microsecond regime. To name a few: electron-electron interaction (thermalization), electron-phonon interaction (electron cooling), intra- and inter-band recombination, as well as quasiparticle diffusion. These dynamical processes happen on ultra-short time scales in the order of femtoseconds to nanoseconds (10^{-15} to 10^{-9}s). Hence, these processes cannot be recorded and probed via convenient electronical detectors and devices, but need to be addressed with special designed optical techniques.
This is done by making use of the unique properties of light with its ability to manipulate or control the material of interest, besides exciting and probing the dynamical processes of desire. In this perspective, the ultrafast laser pulses with time-durations up to the atto-second (10^{-18}s) timescale are used as they are the shortest precisely controllable measurement tools so far, providing the needed time-resolution.
The standard time-resolved spectrocopic method is the pump-probe experiment. Here, two time-locked laser pulses are used. One of the pulses is rather strong and acts as the excitation source, i.e. is used to perturbe the system and excite it to a non-equilibrium state. Then, after certain time-delays, another weaker pulse, the probe beam, is probing the dynamical relaxation process by monitoring the time-dependent change of the optical parameters.

2 Preparation

The objectives for the student are to understand the concepts behind a femtosecond laser system and the principles of nonlinear optics and its applications in current reasearch. As a simple example we measure the transport properties of semiconducting GaAs.
In order to allow for a good understanding of the experiment, it is highly recommended to inform about the following topics and to work on the tasks given in the manuscript before conducting the experiment.

- Be prepared for questions on basic optical components like: prisms, waveplates, polarizers, nonlinear elements (BBO).
- How does a laser system work and what does LASER mean?
- Answer the questions in this tutorial and work on the tasks **before** the day of experiment.
- Understand a fast autocorrelator and its operation principle.
- If questions from your site arise, do not hesitate to contact and ask your supervisor.
- Please wear **clean shoes** at the day of the lab course to prevent the setup of dust.

1 Literature

Here, we give a few books that might be helpful for the preparation of your report. As there are not so many open-source books dealing with the transient grating spectroscopy technique, your supervisor can provide you with additional literature if needed. In addition, you can check the references of the tutorial (especially [1],[2]) and you can even do some research using Google Scholar or arXiv.

- Light-Induced Dynamic Gratings and Photorefraction; H.J.Eichler, A. Hemerschmidt [3]
  `https://www.researchgate.net/publication/225744938_Light-Induced_Dynamic_Gratings_and_Photorefraction`
- Laser Spectroscopy 2, W. Demtröder [4] (available in the data base of the university library)
  `https://link.springer.com/book/10.1007/978-3-662-44641-6#about`
- Light Intensity Modulation Using a PEM (about chopping and lock-in detection)
  `https://www.hindsinstruments.com/wp-content/uploads/Light-Intensity-Modulation-Using-a-PEM-Application-Note.pdf`
- Ultrashort Laser Pulses, C.Antoncini [5] (basics about pulsed lasers)
  `http://www.ull.rdg.ac.uk/documents/ULL_Antoncini.pdf`

3 Basics

1 Time-resolved optical spectroscopy

Traditional optical pump-probe techniques use the unique characteristics of solid-state laser light, which helps to reach time resolution of the desired magnitude and additionally allows to work at very high power levels. A first short laser pulse (pump) in the optical spectrum generates a transient change of the investigated material's optical properties, then a second weaker pulse (probe) monitors the changes. These changes are for instance the change in reflectance, absorbance or transmittance. Measuring the probe beam with various delay in respect to the stimulating beam in time, allows time-resolved studies [6].

Here, we distinguish between two techniques: *Transient Reflectivity* and *Transient Grating*. In the Transient Reflectivity spectroscopy, only two laser pulses are needed as in convenient pump-probe experiments. The material is excited by one pulse and is probed by another. In contrast, in a Transient Grating experiment the material is excited by two pulses which interfere on the materials surface to generate a spatially varying excitation pattern of sinusoidal form. This transient excitation grating is then probed by two weaker probe beams. One beam is specularly reflected, the orther gets diffracted. Simultaneous measurement of both probe beams is called heterodyne detection and allows to increase the signal-to-noise ratio enormously.
In order to be able to understand the details of the Transient Reflectivity and the Transient Grating experiment, we first consider the most important part of the experiment, that are the laser pulses.

2 Characterization of a femtosecond laser system

Here, we describe some basic characteristics of ultra-short laser pulses. Since ultrafast laser pulses are shorter than the smallest time interval that electronic devices can resolve their internal structure, like temporal envelope, cannot be analysed with just a photodetector and an oscilloscope. To wrap up some numbers, currently available high-end oscilloscopes, have a bandwidth of 100 GHz corresponding to an interval of 10 ps while ultrafast laser pulses in the near infrared can be produced with durations of less than 5 fs.
However, there are ways to get at least some information about the characteristics of such short pulses. First, we discuss how ultra short laser pulses can be created and what kind of optical elements as well as principles are required.

2.1 Non-linear optics

In non-linear media the dielectric polarization $\vec{P}$ responds non-linearly to an electrical field $\vec{E}$. In general the polarization can be expressed as

$$\vec{P} = \epsilon_0 \sum_n \chi^{(n)}(\omega) \cdot \vec{E}^n(\omega) = \epsilon_0 \chi^{(1)}(\omega) \cdot \vec{E}(\omega) + \epsilon_0 \chi^{(2)}(\omega) \cdot \vec{E}^2(\omega) + ..., \tag{3.1}$$

where ϵ_0 is the dielectric constant in vacuum and $\chi^{(n)}(\omega)$ is the n-th order dielectric susceptibility. The higher order terms on the right side of equation 3.1 lead to various non-linear optical effects. For example, the second order term can introduce second harmonic generation and Pockels effect, whereas third order terms induce third order harmonic generation and Kerr effects. The details and their theoretical treatment are colorful and rather comlicated, they can be found in many textbooks dealing with nonlinear optics, but are not the focus here.
Note to keep two important aspects in mind: 1) higher order terms (especially $n > 3$) are usually very small and extremely complicated and often not studied in practical. 2) In order to observe second order effects, which are related to rather strict conditions, noncentrosymmetical materials with lack of inversion symmetry are needed. These can be anisotropic crystals.
As an example, we consider the second harmonic generation, which is a process where two photons with the same frequency are absorbed by a material with optically non-linear porperties, and a photon with twice the energy of the initial photons is emitted. So the efficiency for the second harmonic generation is proportional to the square of the electrical field and to the second order suscseptibility $\chi^{(2)}$ (material property). Due to this fact, we use in our setup a BBO (Beta Barium Borate) crystal, because it has a large $\chi^{(2)}$. Later, we exploit this phenomenon and make benefit of it for getting a higher contrast in the autocorrelation measurements, i.e. to increase the signal we measure.

Question: Why is BBO used in the autocorrelation measurement setup? Can you explain the idea and principle of non-linear optics in rather easy words?

2.2 Femtosecond laser

Task: Describe a typical setup for a solid state laser (components, laser gain medium,...). What is needed and how can one understand its working principle (stimulated emission,...)?

In our experimental setup ultra-short pulses are realized by the so called *Kerr-lens mode locking*, i.e. a third order non-linear optical effect is used to modulate the space and time propagation of multiple modes oscillating in the laser cavity. Crucially speaking,the Kerr-effect describes the modification of the refractive index with respect to the electrical field of the light wave. The effect is proportional to the intensity of the laser pulse

$$n(\omega, I) = n_0(\omega) + n_2(\omega) \cdot I(x,t). \tag{3.2}$$

This equation represents the analog of equation 3.1 for the polarizability, so $n_0(\omega)$ is the equilibrium refractive index and $n_2(\omega)$ the second order perturbation in the frequency. To see this and to relate this result to equation 3.1, we write:

$$\vec{P} = [\epsilon_0\chi^{(1)}(\omega) + \epsilon_0\chi^{(3)}(\omega)\vec{E}^2(\omega)]\vec{E}(\omega)] \tag{3.3}$$

In order to bring this into a more convenient form, we formally define an effective linear-like susceptibility χ:

$$\epsilon_0\chi = \epsilon_0\chi^{(1)}(\omega) + \epsilon_0\chi^{(3)}(\omega)\vec{E}^2(\omega) \tag{3.4}$$

and hence equation 3.3 can be rewritten as:

$$\vec{P} = \epsilon_0\chi\vec{E}(\omega) \tag{3.5}$$

Concerning the Maxwell equation $\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0 \varepsilon \vec{E}$ holds and the dielectric constant ε is directly related to the refractive index:

$$n = \sqrt{\varepsilon} = \sqrt{1 + \chi^{(1)}(\omega) + \chi^{(3)}(\omega)\vec{E}^2(\omega)} = \sqrt{n_0^2 + \chi^{(3)}(\omega)\vec{E}^2(\omega)} \tag{3.6}$$

Doing a Taylor expansion of the right side of equation 3.6 in n_0, one ends up with

$$n \approx n_0 + \frac{\chi^{(3)}(\omega)}{2n_0}\vec{E}^2(\omega). \tag{3.7}$$

The transverse intensity distribution of the porpagating light inside a solid state laser can be described by a Gaussian envelope. The beam is therefore more intense at the center then at the edges and according to equation 3.2 a self-focusing effect is induced. The high intensity short pulses including many frequency modes will be focused differently than the continuous wave part. It is therefore desirable to block the continuous waves with lower intensity at the edges by a careful arrangement of an aperture in the beampath. This results in a coherent superposition of the modes in the resonator that survive and is called mode-locking.

Figure 3.1: (a) Kerr-lens mode locking of a Gaussian shaped laser beam. The gain medium/crytsal with refrective index $n(I, \omega)$ is shown in gray. (b) Principle of mode-locking. Each laser has an individual bandwidth dependent on the gain medium (1). The separation of frequencies depends on the cavity length (2) and the resulting amplified modes are shown in (3) [6].

In the cavity of the laser only modes with $\nu_n = n \cdot \frac{c}{2L}$ with $n \in \mathbb{N}$ undergo constructive interference and get amplified as short pulses in the resonator. Locking the relative phase of the modes leads to a strong constructive interference at a given time producing an intense pulse of light circulating within the cavity. The higher the number of modes involved (we use a Ti:Sapphire crystal as gain medium, because it has the capability of providing large enough bandwidth), the shorter the laser pulse can be generated. The relation arising from Fourier relations between duration τ_{Pulse} and spectral width $\Delta\nu$ is given by the so-called time-bandwidth product

$$\Delta\nu \cdot \tau_{\text{Pulse}} \geq K. \tag{3.8}$$

Here, K is a constant dependent on the pulse's envelope and describes the minimum achievable pulse duration (FWHM intensity) at a given bandwidth $\Delta\nu$. If equality holds, the pulse is called Fourier-limited and has a theoretical minimal pulse duration.

Question: What are the values K for a (1) Gaussian and (2) Lorentzian shaped laser envelope?
Question: Discuss the time-bandwidth product. What are the pros and cons for ultrashort pulses in experiments (take equation 3.8 into account)? Can you imagine reasons why the pulses in reality are somehow a bit stretched in time?

4 Setup

1 Interferometric second order autocorrelation

In this chapter we demonstrate the working principle of our home-built optical autocorrelator. The aim is to measure the pulse duration defined as the Full Width at Half Maximum (FWHM) of the electric field envelope in time domain.
It is absolutely necessary for a good interpretation of measurements to know the pulses conditions and physical parameters (e.g. temporal duration, spectrum,...). As discussed above, measuring a short event is usually achieved under the use of even shorter ones. It is clear that one uses the technically feasible shortest beam for the experimental measurement, so there is no shorter pulse available and logically the pulse has to be measured somehow by itself. The corresponding technique is known as *optical autocorrelation*.

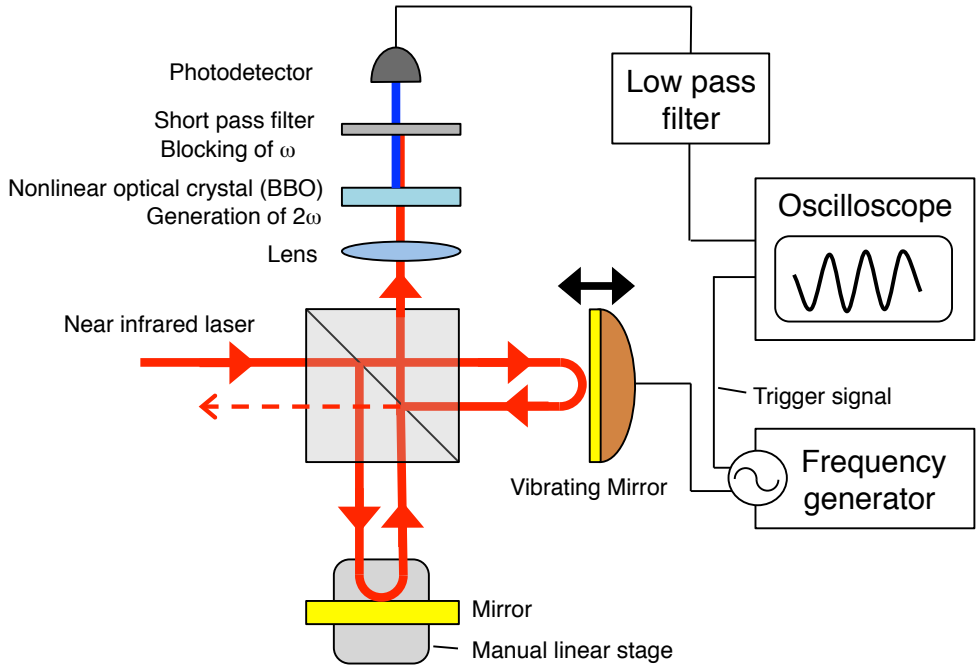


Figure 4.1: Scheme of the setup to measure the autocorrelation of a laser pulse. Optical components are labeled in the figure. For details see main text.

In the autocorrelator each incoming pulse is split by a beam splitter into two identical copies that are sent along two different paths of the arms of a standard Michelson interferometer. In the Michelson interferometer one mirror is spatially fixed but connected to a vibrating stage coupled to a radio frequency generator. In this way, the vibrating mirror oscillates with a certain frequency around its equilibrium position. This oscillation causes an oscillating time difference of $\pm\tau$, so the one beam will be shifted in time ($\tilde{t} = t \pm \tau$) with respect to the other one. Crucially speaking, the two beams interfere constructively or destructively, depending on the induced delay. The other mirror can be moved back

and forth and allows to change the optical distance between the two pulses and to find the equilibrium distance that corresponds to $\tilde{t} = t$. Behind the intereferometer the pulses are recombined collinear in space by adjusting the associated optical elements and focused on a nonlinear crystal (BBO for Beta Barium Borate). Passing the BBO crystal the pulses frequency is doubled, i.e. second harmonic generation ($E_\omega \rightarrow E_{2\omega}$). The outgoing light is then focused onto a photodiode connected to an oscilloscope to detect the autocorrelation signal. The interferometric configuration allows to measure the combined quantities of the normal intensity autocorrelation and the spectrum (fringe-resolved structure) in a single data trace.

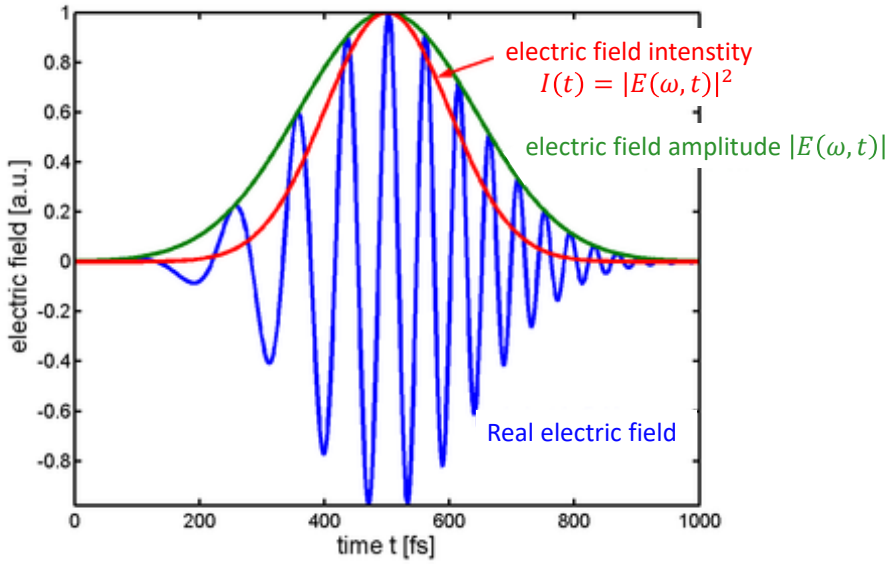


Figure 4.2: The different quantities of an oscillating electric field $E(\omega, t)$.

Question: Why is it necessary to make the two pulses overlap both in space and time and how can this be realized?
Question: Can you imagine why it is beneficial to make use of the second harmonic generation, i.e. use $E_{2\omega}$ rather than E_ω? (hint: think about the measured quantity)

Quantitative analysis

Requirements for a good autocorrelation measurement are the knowledge about the temporal shape of the laser pulse and about the carrier frequency (central frequency of the laser irradiation). Both of them can be determined, since we can approximate the output of the pulsed laser by a Gaussian pulse and the carrier frequency can easily be measured with a spectrometer [5].

The measured quantity from the photodiode, that is the autocorrelation signal $I_{AC}(\tau)$, can be expressed as:

$$\begin{aligned} I_{AC}(\tau) &\propto \int_{-\infty}^{\infty} \left[|E(t) + E(t-\tau)|^2\right]^2 \mathrm{d}t \\ &= \int_{-\infty}^{\infty} |E^2(t) + 2E(t)E(t-\tau) + E^2(t-\tau)|^2 \mathrm{d}t \end{aligned} \tag{4.1}$$

Here, $E(t)$ and $E(t-\tau)$ are the electric field amplitudes of the two pulses and τ is the time delay introduced by the vibrating mirror.
From equation 4.1 it can be seen that there are many contributions to the autocorrelation signal arising. In order to analyse the obtained signal properly, it is necessary to take into account all the different terms.

$$I_{AC}(\tau) = \int_{-\infty}^{\infty} \{I^2(t) + I^2(t-\tau)\}\mathrm{d}t + \int_{-\infty}^{\infty} \{I(t) + I(t-\tau)\}\mathrm{Re}\{E(t)E^*(t-\tau\}\mathrm{d}t + \int_{-\infty}^{\infty} \mathrm{Re}\{E^2(t)E^*(t-\tau)^2\}\ \mathrm{d}t + \int_{-\infty}^{\infty} I(t)I(t-\tau)\mathrm{d}t \tag{4.2}$$

In equation 4.2 the first term on the right side represents the second harmonic generation intensity (constant), the second term corresponds to a modified intereferogram of the field $E(t)$, the third term is the interferogram of the harmonic frequency field and the last term represents the intensity autocorrelation.
In order to determine the temporal width of the laser pulse, it is sufficient to consider the envelope of the autocorrelation signal.

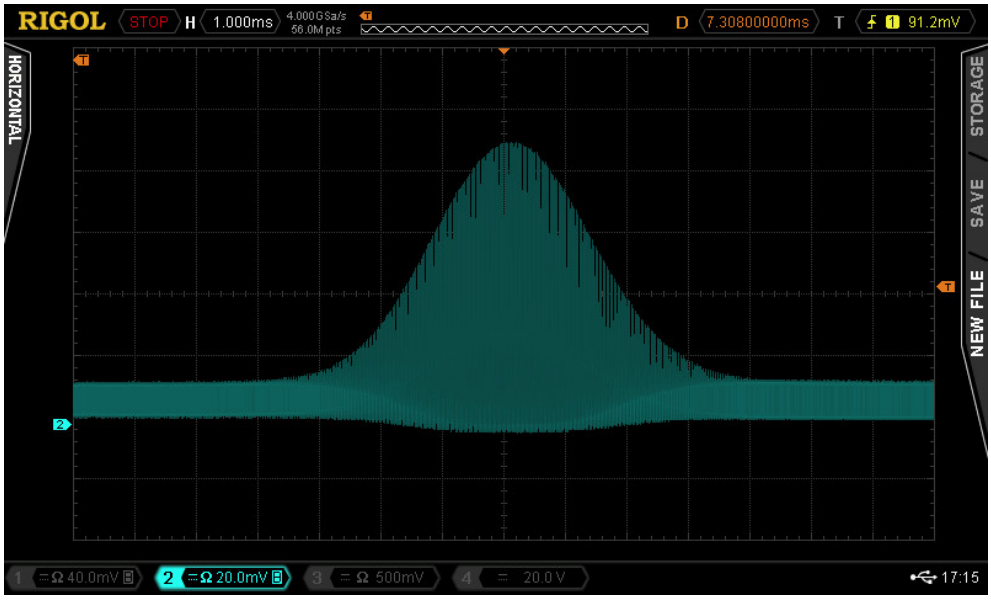


Figure 4.3: Autocorrelation signal at the oscilloscope.

Question: Can you explain why it is for the determination of the laser pulse duration sufficient to study the envelope of the autocorrelation function rather than all details?
Question: Assume the field $E(t)$ can be expressed as a Gaussian $E(t) = e^{-t^2/\Gamma^2}$, with Γ the Full Width at Half Maximum. How to relate the obtained Γ to the real pulse duration? What do we actually measure?

2 Optical pump-probe spectroscopy

After establishing the basic principles about ultra-short laser pulses, we can now turn to discuss the two pump-probe experiments, the Transient Relectivity and Transient Grating

technique, in detail.

Optical pump probe spectroscopy is a powerful tool for the study of non-equilibrium properties of a material. It exploits the fact that most parameters of a material couple to its optical properties. In optical pump probe spectroscopy a deviation from thermal equilibrium is laser-induced. This transient pertubation leads for instance to a change of the sample's complex reflectivity r.
In a first step a strong laser pulse (pump) perturbs the material, then a second pulse (probe) monitors the resulting transient changes of the physical properties. By repeating the measurement with different delays between the pump and the probe pulse, time traces of the excitations are acquired. (In other words: you take a 'photo' at different time delays, stick them together and you get a continuous image (like a movie) of the dynamics.) Their evolutions reveal information about the relaxation processes and the responsible material properties. Therefore, the study of carrier generation, recombination and transport processes, including the determination of carrier lifetimes and their dynamics is feasible [6].
However, the standard pump-probe experiment can only measure the average global excitation and is hence only sensitive to recombination processes as it lacks spatial resolution. In order to resolve also spatial dynamics, like diffusion of the excited quasiparticles, one has to make an extension of the simple pump-probe technique. This is done by the Transient Grating technique in which a spatial modulation to the pump induced excitation pattern is added. This allows to distinguish between local and nonlocal contribution to the occuring dynamics and relaxation [1, 6].

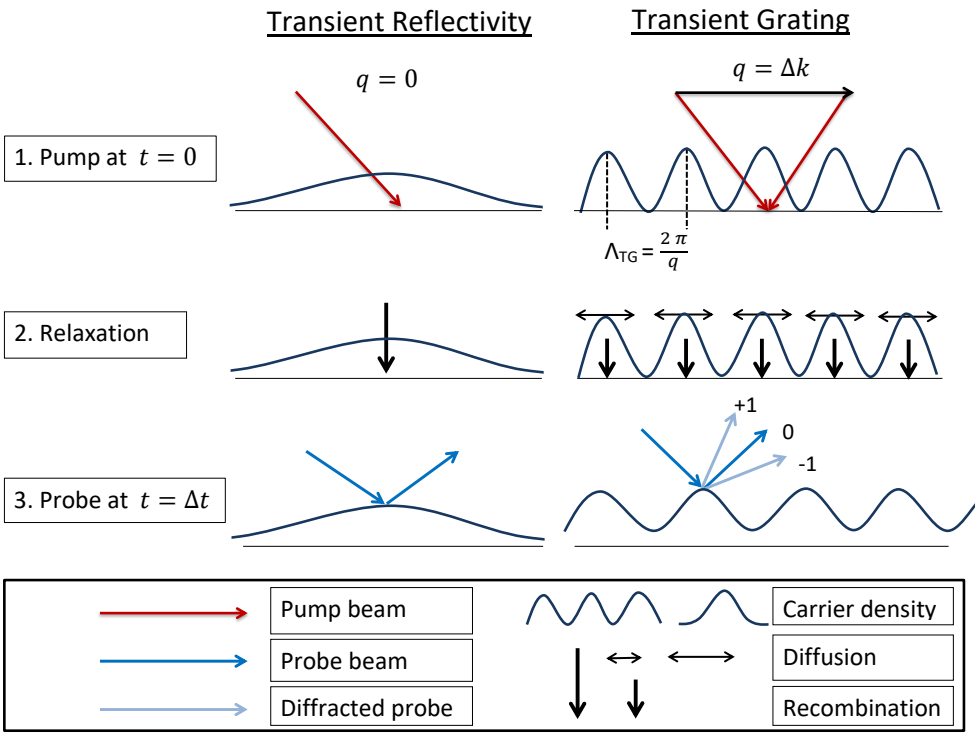


Figure 4.4: Principle scheme of Transient Reflectivity (left) and Transient Grating (right) spectroscopy, if just recombination and diffusion are taken into account. The excitation, relaxation and detection of free carrier population is illustrated.

2.1 Transient Grating spectroscopy

In the Transient Grating spectroscopy two time coincident non-collinear pump laser beams interfere at the sample surface and create a sinusoidal modulated intensity pattern. This spatially modulated excitation pattern has a period Λ_{TG}. The grating period given by $\Lambda_{TG} = \lambda/2\sin(\theta/2)$ is determined by the laser wavelength λ and the angle θ between the incident non-collinear pump beams measured relative to the sample surface normal [7]. This additional spatial resolution connected to the period of the laser induced optical grating allows for the simultaneous determination and distinction of recombinative and diffusive relaxation mechanisms of the excitation population.

The Transient Grating principle:

- The two pumps produce an intensity grating, i.e. standing wave of varying intensity on the surface due to their interference.
- The material responds to this present standing wave of intensity. An excitation grating emerges.
- The spatially modulated excitation density changes the optical properties of the material. A Transient Diffraction Grating emerges.
- The probe beams test this Transient Grating as they once scatter off the optical grating (diffraction) and get specularly reflected.
- The optical grating relaxes locally (depopulation/recombination) and non locally (diffusion).
- The time delay between the pump and probe beams is changed continously and the time evolution of the Transient Grating is detected.

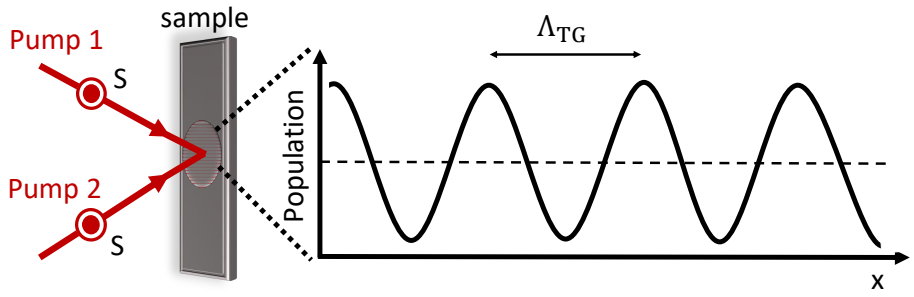


Figure 4.5: General scheme of the Transient Grating creation by interference of two co-linear polarized pump beams. The intensity grating is shown on the right which modulates the excitation density in real space x and hence modifies the optical properties.

The Transient Grating technique is a four-wave-mixing process and in terms of non-linear optics is related to the third-ordner non-linear susceptibility $\chi^{(3)}(\omega)$ [8].

Evolution of a Transient Grating

The laser induced spatially modulated intensity grating leads to a local imbalance, i.e. a gradient of the excitation density in real space. This imbalance causes diffusion of the excited particles along this gradient from areas of high excitation densities towards the lower ones. In order to find the temporal evolution of a sinusoidal excitation pattern $n(x,t)$ like a Transient Grating, it is sufficient to solve a one dimensional diffusion equation that also includes an additional term considering the overall recombination dynamics.

$$\frac{\partial}{\partial t}n(x,t) = D_a \frac{\partial^2}{\partial x^2}n(x,t) - \frac{1}{\tau_{\text{rec}}}n(x,t) \tag{4.3}$$

Here, the partial derivative is taken with respect to x, which is the spatial coordinate perpendicular to the grating fringes. D_a is the ambipolar diffusion coefficient and τ_{rec} is the recombination time. Since this is a linear equation, it can be solved in Fourier space.

$$\frac{\partial}{\partial t}n_q = -(D_a q^2 + \frac{1}{\tau_{\text{rec}}})n_q \tag{4.4}$$

The spatial frequency or wave number $q = 2\pi/\Lambda_{\text{TG}}$ of the Transient Grating is given by the fringe spacing Λ_{TG} (wavelength of the modulation).

Task: Show that $n(x,t) = n_0\{\exp(-t/\tau_{\text{rec}}) + \cos(qx)\cdot\exp(-(q^2 D_a + \frac{1}{\tau_{\text{rec}}})t)\}$ solves equation 4.3.

Task: Do the Fourier transformation of equation 4.3 to get a better feeling of what is happening there.

Question: What is the lifetime of the q-component of excess carrier distributions with:
- a) a finite value of q?
- b) $q = 0$? To which experimental situation does this case correspond?
- c) What has to be done experimentally to solve for the diffusion coefficient D_a?

Bonus: Can you also imagine other kinds of laser generated Transient Gratings besides the here presented excitation density grating? Discuss in your group.

Detection of the Transient Grating dynamics

In convenient pump-probe experiments, which rely on so-called *homodyne* detection, only one probe beam is used to record the temporal evolution of the laser induced excitation. In this way, the intensity $I(t) \propto |E(t)|^2$ of the reflected probe pulse is measured by the photodiode. This is often not beneficial as the intensity signal is very weak and the changes in the optical parameters, for example reflectivity $r \to r_0 + \delta r$ with $|\delta r/r_0| \sim 10^{-4}$, are very small. To overcome this issue, we *heterodyne* our probing signal. In heterodyne detection one uses two probe beams to record the transient evolution of the excitation. The second probe beam is called local oscillator (LO). Due to a special alignment configuration, called BOXCAR geometry [9], the two probe beams are aligned in such a way that they travel collinearly to the detector after interaction with the Transient Grating. Here, one probe beam (LO) gets specularly reflected (as in the transient reflectivity) and the second probe gets diffracted by the Transient Grating as shown in Fig. 4.4.
In this way, the recorded heterodyne intensity signal is:

$$\begin{aligned} I_{\text{het}} &\propto |E_{\text{Dif}}(t) + E_{\text{LO}}(t)|^2 \\ &= |E_{\text{Dif}}(t)|^2 + |E_{\text{LO}}(t)|^2 + 2|E_{\text{Dif}}(t)||E_{\text{LO}}(t)|\cos(\Psi) \end{aligned} \tag{4.5}$$

In equation 4.5 the first term $|E_{\text{Dif}}(t)|^2$ corresponds to the homodyne term, which is small and at the egde of the accessible resolution of the detection devices. The second term $|E_{\text{LO}}(t)|^2$ contains information about the average excitation changes and is the transient reflectivity term. It can be easily substracted by lock-in detection. The important term is the last one as it is rather proportional to the amplitude of the electric field than to its intensity. Therefore, it is large compared to the homodyne term and preserves many orders of magnitude in the signal. In addition this term can be periodically modulated by controlling the heterodyne phase Ψ. In our setup the phase Ψ is precisely controlled by putting a thin glass plate into the beampath of the LO. Due to the combination of lock-in detection and the heterodyne probing scheme the heterodyne signal that is measured can be expressed as:

$$S_{\text{Het}} \propto S_{\text{Pop}} + S_{\text{TG}} \cdot \cos(\Psi) \tag{4.6}$$

Hence, it is possible to measure the transient reflectivity which is encoded in the S_{Pop} signal and the evolution of the Transient Grating S_{TG} simultaneously. In order to do so, at each certain time delay the heterodyne phase is modulated by rotating the glass plate (=coverslip) by a small finite angle. The principle detection scheme and the resulting data trace at one ceratin time delay is shown in Fig. 4.6.

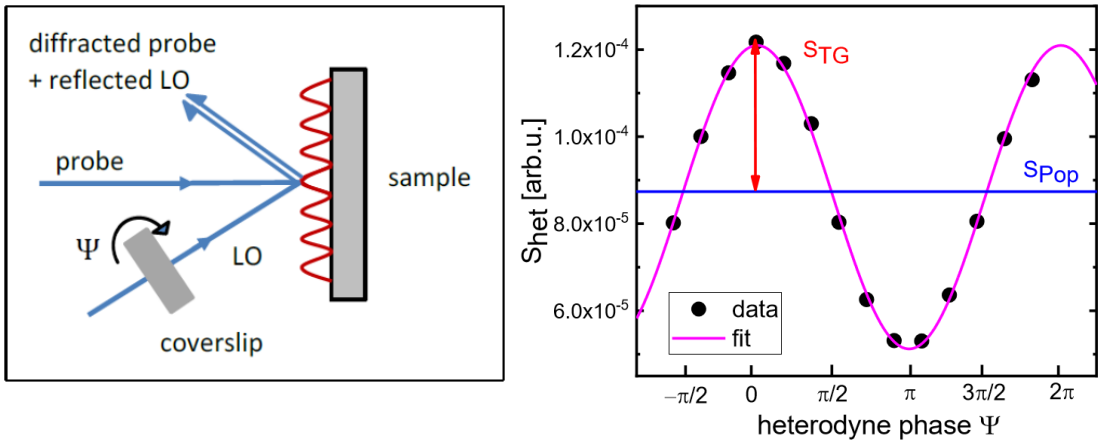


Figure 4.6: General scheme of the Transient Grating detection by heterodyning the reflected Lo and diffracted probe beam. For clarity only the probe beams (blue) are shwon. The obtained heterodyne signal is shown on the right. It can be clearly seen that the heterodyne signal as a function of the heterodyne phase Ψ is periodically modulated. The data can be fitted with an offset-cosine function.

Finally, we point out that the Transient Reflectivity part S_{Pop} illustrates only the recombination dynamics, whereas the Transient Grating part of the signal S_{TG} decays due to the recombination and additionally due to the diffusion processes. This means we can determine the recombination rate $k_{\text{rec}} = \sum_{i=1}^{\infty} k_{\text{rec},i}$ and the ambipolar diffusion rate $k_{\text{Dif}} = D_a q^2$. Hence, we can infer:

$$
\begin{aligned}
S_{\mathrm{Pop}} &\propto \sum_{i=1}^{\infty} \exp\left(-k_{\mathrm{rec},i} t\right) \\
S_{\mathrm{TG}} &\propto \sum_{i=1}^{\infty} \exp\left(-k_{\mathrm{rec},i} t\right) \exp\left(-D_a q^2 t\right)
\end{aligned}
\tag{4.7}
$$

Because the thermalization of electrons happens on a short time-scale ($\approx 100\ fs$), which we can not observe, we can only study the recombination dynamics (e.g. interaction of electrons with phonons,...) and the ambipolar diffusion.

Question: Why is the heterodyne detection scheme so important?
Question: What has to be done experimentally to extract and separate S_{Pop} and S_{TG} from the heterodyne signal S_{Het}?
Question: What has to be done experimentally to determine the recombination rate and the diffusion rate? Are there different possibilities?

Finally, we present a schematic sketch of the experimental Transient Grating setup shown in Fig. 4.7. After splitting the pulse into the pump and probe beam and change their relative polarization in order to prevent the detector from scattered pump light (detector measures only polarization of probe beam). The pump beam is chopped so the lock-in-amplifier will select the signal with the frequency given by the chopper and the noise will be reduced.

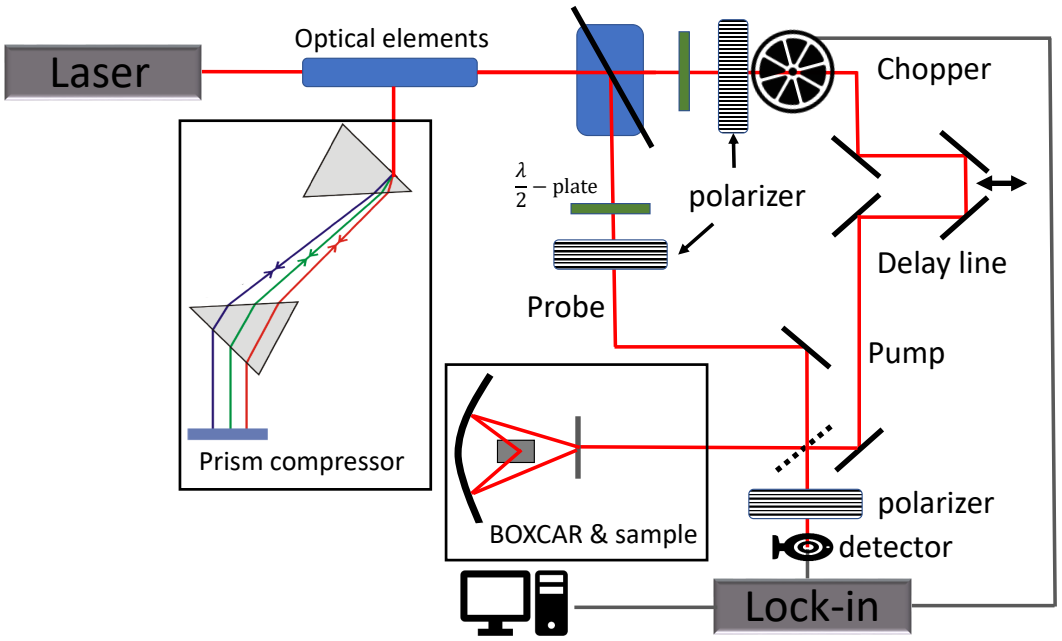


Figure 4.7: Schematics of the setup for Transient Grating and Transient Reflectivity measurements.

Question: Explain, in a few sentences, how a chopper and a lock-in-amplifier works and what the advantages of a prism compressor are.

3 Photoexcitation and relaxation

The absorption of an indcident photon with an energy of at least the size of the band gap of the investigated material can excite an electron-hole pair in the bandstructure or leads to the emergence of other qausiparticles. This photo-excited state can relax via several processes: electron-electron interaction (thermalization), electron-phonon interaction (electron cooling), intra- and interband recombinations and ambipolar diffusion. In this section we want to introduce you to these processes.

Thermalization and electron cooling

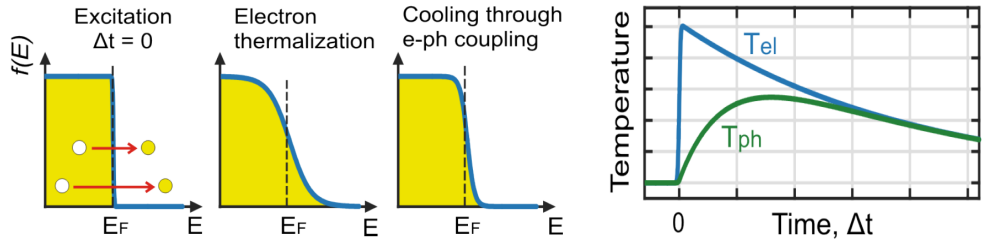


Figure 4.8: Left sketch shows electronic and lattice temperature relaxation which causes the energy redistribution of electrons for various time delay to change. Right panel shows an illustration of electronic and lattice temperature relaxation in a metal after excitation with an optical pulse [1].

The optical pump pulse perturbs the sample by direct coupling to the electrons in the sample and initially generates particle excitations across the electronic gap in case of a semiconductor. These excitations quickly thermalize amongst each other within $\sim$ 100 fs. The pump pulse can raise the effective electronic temperature T_{el} rapidly, so the electrons follow a Fermi-Dirac distribution with another temperature, depending on the excess energy (see Fig. 4.8). Then the electronic system transfers some of this energy to the lattice through electron-phonon coupling. In this way the temperature of the lattice T_{lat} increases. With time T_{lat} and T_{el} equilibrate with typical timescales of few picoseconds.

Ambipolar diffusivity

Optical excitation of a carrier population that is spatially non-uniform initiates a diffusion process, driven by the gradients of the densities of excited electrons $n(x,t)$ and holes $h(x,t)$ (*ambipolar*: diffusion of positive and negative carrier species with opposite electrical charge due to their interaction via an electric field). Coulomb interaction couples the diffusion of the two carrier species so that

$$n(x,t) \approx h(x,t). \tag{4.8}$$

This interaction can be taken into account by an additional drift term in Fick's first law [10], describing the particle currents $\vec{j}_e$ and $\vec{j}_h$ of electrons and holes respectively

$$\vec{j}_e = -D_e \nabla n - \mu_e n \vec{E}(x,t) \tag{4.9a}$$

$$\vec{j}_h = -D_h \nabla h + \mu_h h \vec{E}(x,t) \tag{4.9b}$$

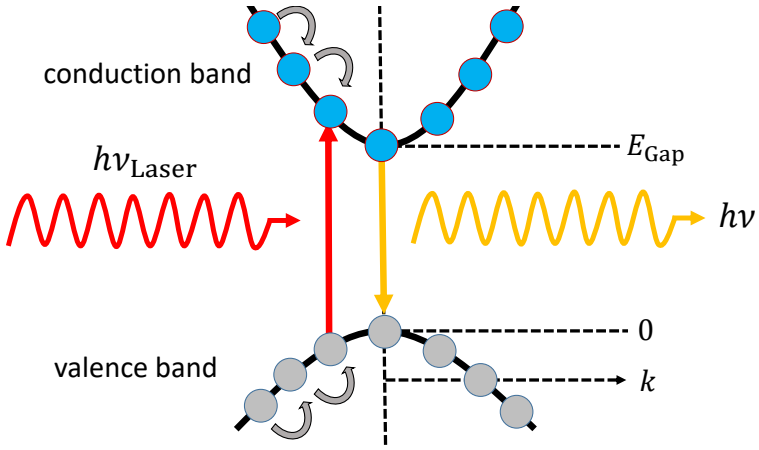


Figure 4.9: After photoexcitation with a laser of energy $h\nu_{\text{Laser}}$ carriers make a transitions from states in the valence band to states in the conduction band. Then, the excited carriers realx to the minimum of the band by various interaction processes like electron-phonon coupling, until the created electron-hole pairs make an interband recombination and emit a photon with the energy of the gap. This photon can be observed in photoluminescence measurements.

with D_e, D_h, μ_e and μ_h being the diffusion coefficients and mobilities of the two respective species and $\vec{E}$ is the electric field caused by two different charges which are inhomogeneous distributed. Eliminating $\vec{E}$ and using (4.8) and $\vec{j}_e \approx \vec{j}_h =: \vec{j}$ leads to

$$\vec{j} = -\frac{D_e\mu_h + D_h\mu_e}{\mu_h + \mu_e} \cdot \nabla n. \tag{4.10}$$

The comparison of (4.10) with Fick's first law allows for the identification of the ambipolar diffusivity

$$D_a = \frac{D_e\mu_h + D_h\mu_e}{\mu_h + \mu_e} = \frac{k_B T}{e} \cdot \frac{2\mu_h\mu_e}{\mu_h + \mu_e}. \tag{4.11}$$

For the right side of (4.11) the Einstein relation for both species $i = e, h$

$$D_i = \mu_i \frac{k_B T}{e}, \tag{4.12}$$

with the Boltzmann constant k_B, the temperature T and the elementary charge e was used. An ambipolar mobility for this combined system can be defined by comparing (4.11) and (4.12):

$$\mu_a = \frac{2\mu_h\mu_e}{\mu_h + \mu_e}. \tag{4.13}$$

In equation (4.13) one can distinguish two limiting cases:

$$(1) \qquad \mu_h = \mu_e \qquad \Rightarrow \qquad \mu_a = \mu_h = \mu_e \tag{4.14a}$$

$$(2) \qquad \mu_j \gg \mu_i \qquad \Rightarrow \qquad \mu_i = 2\mu_j \ \textit{with} \ \ i, j \in \{h, e\}. \tag{4.14c}$$

Question: Interpret the different cases. What property of the band structure can influence the mobility of the carriers?

5 Experimental procedure

In general: Take care only to touch the optics that you are supposed to operate!
The experiment is split into two parts.

First, you will complete the optical setup for the fast autocorrelator and measure the optical pulse width. Here, you need to align the two pulses traveling in the Michelson interferometer properly. If the time allows, you will measure the autocorrelation before and after passing the prism pulse compressor. From the measured spectrum of the pulses you can also determine the spectral composition of the laser radiation.

In the second part of the experiment, you will conduct a heterodyned Transient Grating experiment on GaAs together with your tutor.

6 Analysis

- Calibrate and rescale the x-axis of the autocorrelation signal traces as explained by your tutor.
- Determine the pulse duration (intensity FWHM) from the autocorrelation measurement, as well as the bandwidth of the spectrum. Compare and discuss the resulting products to the theoretical transform limit. What can cause deviations?
- Extract the electron cooling time-constant of hot carriers in GaAs (and compute the average electron-phonon-coupling coefficient).
- Determine the decay times for the Transient Reflectivity and Transient Grating signals. Deduce the recombination time as well as ambipolar diffusivity and mobility in GaAs. Discuss your result and compare to literature (search for scientific puplications).

Try to explain the demanded principles in your own words. Please, try to keep your report simple and short. Keep in mind that you write for yourself and you need to understand what you wrote down.

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